

Custard — Crème Anglaise

Cream and milk, warmed. Yolks and sugar, whisked pale. Temper. Back to the pan. Stir constantly to 82°C — not 83, not 85. Strain. Ice bath. The base every serious dessert on a menu leans on.

Yield: **750 mL** (8) Active: **25 min** Total: **45 min** + chill Target: **82°C** (180°F) Window: **80–84°C**

INGREDIENTS

METRIC · VOLUME RIGHT COLUMN

CRÈME ANGLAISE BASE

Whole milk	250mL	1 cup
Heavy cream (35%+)	250mL	1 cup
Large egg yolks (<i>no whites</i>)	6	~110g
Granulated sugar	100g	½ cup
Fine sea salt	1g	¼ tsp
Vanilla bean (<i>split + seeds</i>)	1	—

PASTRY CREAM VARIATION

Whole milk	500mL	2 cups
Egg yolks	4	~75g
Sugar	100g	½ cup
Cornstarch (<i>not flour</i>)	40g	½ cup
Salt	1g	¼ tsp
Vanilla	1 pod	—
Butter (<i>off heat</i>)	30g	2 tbsp

FLAVOUR VARIATIONS

Coffee	20g ground, infuse warm 15 min, strain
Chocolate	hot anglaise over 150g 70% choc, whisk
Praliné / gianduja	100g paste, whisk in warm
Cardamom	8 crushed pods / 2 tbsp tea, infuse, strain / tea
Citrus	zest in dairy (NEVER juice — curdles)
Rum / Cognac	20–30 mL cold, at service
Saffron	10 threads bloomed in 15 mL hot water
Brown butter	replace 100 mL cream w/ brown butter

Kit: heavy 3 L saucepan (light interior) · silicone spatula · *instant-read thermometer* · chinois · ice-bath bowl · whisk (yolks only)

METHOD — CRÈME ANGLAISE

- Infuse dairy.** Milk + cream + split vanilla (seeds + pod). Bare simmer, cover, steep **10 min**. Coffee / tea / zest / saffron infuse now.
Vanilla volatiles flee a rolling boil; a gentle steep extracts without damage.
 - Yolks to ribbon.** Yolks + sugar + salt. Whisk vigorously **2–3 min** until pale and ribboning.
Sugar coats the proteins and raises their coagulation temp by 10–15°C. Under-whisked = pre-scrambled. — McGee, pp. 97-98
 - Temper.** Re-warm dairy to steam. Pour 1/3 into yolks in a thin stream, whisking constantly. Return tempered yolks to the pan with remaining dairy. Switch from whisk to silicone spatula.
Gradual climb lets proteins unfold rather than seize. Every curdled custard is either un-tempered or too hot.
 - The stir.** Medium-low heat. **Figure-eight with corner sweeps**, continuously. No whisking — it aerates and hides the texture. 5–8 min.
Coagulation danger is at the pan bottom and corners where heat concentrates. Figure-8 means every cm² is scraped every 2–3 sec.
 - Read the thickening** at ~75°C: clean trail on the pan bottom, surface goes glossy. Two to three more minutes.
- Target 82°C (180°F).** Nappe test: line held across the spatula back with a fingertip. Thermometer in the liquid, not touching pan.
80°C = thin. 82°C = perfect. 84°C = edge. 85°C = curdled. No recovery above the ceiling.
- THE STOP.** At 82°C, off heat *immediately*, pour through chinois into the ice-bath bowl. Do not stop at the heat — carryover hits 85°C in seconds.
- Stir over ice** 1–2 min to drop under 30°C. Press plastic wrap directly on the surface. Chill **2 hr minimum**, ideally overnight — flavour develops.

Pastry cream variant: starch in yolks + whisk (not stir) + **BOIL 90 sec** (amylase kill — unboiled pastry cream liquefies overnight). Off heat, 30 g butter. Film on surface. Whisk before use.

Sous-vide variant (bulletproof): same ingredients → bag → 82°C / 45 min / agitate every 15 min → chinois → ice bath. Failure rate zero.

ELEVATION

TIER 1 — NO EXTRA TIME

- Tahitian vanilla** — floral/cherry-cola vs. Bourbon
- Brown sugar swap** — 30 g of total
- Extra yolk** — 7 for ice-cream base
- Malt syrup** — 10 g, Horlicks register

TIER 2 — WORTH IT

- Sous-vide** — 82°C / 45 min / unfaillable
- Caramelised cream** — dry-caramel 40 g, deglaze off heat
- Brown-butter base** — replace 100 mL cream
- Duck yolks** — 4 ducks for 6 hens, richer
- Crème brûlée** — same base, ramekins, bain-marie 150°C 30–40 min, brûlée at service

TIER 3 — RESTAURANT

- Ice cream** — 7 yolks + 60 g brown butter, churn, harden 4 hr
- Pot de crème** — espresso-cup bake, bain 150°C 25–35 min
- Custard espuma** — iSi + 1 N₂O, chill 1 hr
- Île flottante** — anglaise + poached meringue + caramel
- Sabayon** — Marsala replaces 100 mL cream, bain-marie whisk

TROUBLESHOOT

Curdled / grainy	Past 84°C. Stick-blender over ice bath (saves 70%); strain hard.
Too thin chilled	Under-cooked (<80°C). Use as thin sauce; re-cook impossible.
Too thick	Over-reduced. Thin with warm cream, spoonful at a time.
Eggy smell / sulphur	Over-cooked. 82°C target, chinois, non-negotiable.
Skin on surface	No film contact. Press plastic directly on custard.
Pastry cream soup AM	Amylase not killed. Must boil 90 sec whisking.
Ice cream icy	Not enough fat / no alcohol. 20 mL vodka per L base.
Brûlée top soft	Too little sugar / damp top. 2–3 mm sugar, pat dry, 2–3 torch passes.

CHARTER PREP & STORAGE

Galley move	Sous-vide 1 L → 3 desserts: sauce, ice cream, pot de crème.
Anglaise	3 d fridge. Never freeze — breaks. Churn to ice cream instead.
Pastry cream	3 d fridge max. Film on surface. Never freeze — starch retrogrades.
Sous-vide bag sealed	1 wk fridge unopened.
Ice cream churned	2 mo freezer. 20 mL vodka / L prevents over-hardening.
Pot de crème baked	3 d fridge sealed.
Scaling	Stovetop cap at 1.5 L / pot (corner dead zones). Sous-vide scales to any volume.
Target never changes	82°C . Never adjust for batch size.